Chapter 7: Software Quality

Assurance & Testing

Ensuring Reliability, Correctness,

and Performance

Focus: From "It works on my

machine" to "It works for the user."

Learning Objectives

• Overview: What is testing and why do we do it?

• Concepts: Failures, Faults, and Errors.

• Activities: The levels of testing (Unit to Acceptance).

• Managing Testing: Plans and Roles.

• OO Impact: How classes and objects change testing

strategies.

• Types: Black-box vs. White-box testing.

7.1 An Overview of Testing

• Verification vs. Validation

• Software Quality Assurance (SQA): The systematic process

of checking if software meets specified requirements.

• Verification: "Are we building the product right?" (Checking

code against design).

• Validation: "Are we building the right product?" (Checking

software against user needs).

• Example:

• Verification: A developer reviews the code for the

calculateGPA() function to ensure it follows the math

formula correctly.

• Validation: We show the "Grade Report" screen to the Registrar to ask, "Is this the layout and information you need to print transcripts?"

7.2 Testing Concepts: The Chain of Defects

Reliability Terminology It is crucial to distinguish between the cause and the effect of a bug.

• Fault (Bug): The mechanical/algorithmic defect in the code.

• Error: The incorrect internal state of the system caused by the fault.

• Failure: The external, visible deviation from expected behavior.

Example:

• Fault: The programmer types if (capacity < 0) instead of if (capacity == 0) in the logic.

• Error: The variable isFull remains false even when the course has 0 seats left.

• Failure: The system allows a 31st student to register for a 30-seat class.

7.2 Testing Concepts: Test Components

Tools of the Trade

• Test Case: A set of inputs, execution preconditions, and expected outcomes.

• Test Stub: A dummy piece of code that simulates a component that is called by the code being tested.

• Test Driver: A dummy piece of code that calls the code being tested.

Example:

• We are testing the CourseManager class, but the Database isn't built yet.

• Stub: We write a FakeDatabase that just returns "Success" whenever CourseManager tries to save data.

• Driver: We write a simple script MainTest.java that manually calls CourseManager.addStudent("John") to see if it works.

7.3 Testing Activities (The V-Model)

Levels of Testing

• Unit Testing: Testing individual components (classes/methods) in isolation.

• Integration Testing: Testing how those components interact (interfaces).

• System Testing: Testing the entire system as a whole against requirements.

• Acceptance Testing: The client tests the system to decide if they will pay for it.

Example:

• Unit: Testing the Student.getName() method.

• Integration: Testing if Student object can successfully be saved into the Course list.

• System:Testing the full "Register for Semester" workflow (Login ->